1.3 Adding in binary

In this lesson

You will learn:

- how to do simple addition in binary
- what overflow means when adding in binary.

When digital data is stored as a value it can be used in calculations. For example, you have used spreadsheets in this course. When you enter a formula such as =A3+B3 in a spreadsheet cell your computer will carry out binary addition. You have also learned programming in this course. When your computer moves a sprite on screen it uses binary addition to calculate a new position.

In this lesson you will learn how to do simple addition using binary.

Simple addition in decimal

To do addition in binary you use the same method as addition in decimal. It will help to look at an example of how addition works in decimal before moving on to binary addition.

If you lay out the addition in a table like the one on the right, it is easier to follow what happens when you add the numbers. You use the first two rows in the table for the numbers you will add together. You use the row at the bottom to record the sum. You use the shaded row to hold any values that you need to carry.

Number 1			
Number 2			
Carry			
Sum			

Example: decimal addition

In this example you will add together the numbers 262 and 174.

When you do an addition, you add together the numbers in each column from right to left and record the sum.

Step 1: Add the Units column: Add 2 and 4 together. Think of the sum as being '06'. This will help when you do binary addition later.

Write '06' in the table. Enter the 0 in the carry row of the 10s column. Enter the 6 in the sum row of the Units column.

	100s	10s	Units
Number 1	2	6	2
Number 2	1	7	4
Carry		0	
Sum			**6**

Step 2: Add the 10s column: The digits in this column add up to 13. Enter the 1 in the carry row of the 100s column. Enter the 3 in the sum row of the 10s column.

	100s	10s	Units
Number 1	2	6	2
Number 2	1	7	4
Carry	1	0	
Sum		**3**	**6**

Step 3: Add the 100s column: The digits in this column add up to 04. There is nothing to carry. Enter the 4 in the sum row of the 100s column. This completes the addition.

	100s	10s	Units
Number 1	2	6	2
Number 2	1	7	4
Carry	1	0	
Sum	**4**	**3**	**6**

262 + 174 = 436.

Activity

Draw a copy of the table used in the previous example. Use it to add 729 and 252.

Adding in binary

You use the same method to add two binary numbers together. Adding in binary seems more difficult because you are not so familiar with binary numbers. There are four rules that will help you to do binary addition.

Rule 1: 0 + 0 = **00**

Number 1		0
Number 2		0
Carry	**0**	**0**
Sum		**0**

Rule 2: 0 + 1 = **01**

Number 1		0
Number 2		1
Carry	**0**	**0**
Sum		**1**

Rule 3: 1 + 1 = **10**

Number 1		1
Number 2		1
Carry	**1**	**0**
Sum		**0**

Rule 4: 1 + 1 + 1 = **11**

Number 1		1
Number 2		1
Carry	**1**	**1**
Sum		**1**

You can use these rules to add any two binary numbers. Write the two numbers one above the other so the bits line up. Start with the bits in the Units column (on the right). Look down the column. What bits do you see? It will be one of these possibilities:

- 0 + 0
- 1 + 0
- 1 + 1

The rules of binary addition will tell you the answer. Write in the answer and any carry bit.

Now go to the next column (the 2s column). Look down the column, including any carry bit. What bits do you see? The same possibilities as before. Because of the carry bit there is another possibility:

- 1 + 1 + 1

Write in the answer plus any carry bit. Do the same for every column until you have added every column.

You will see an example on the next page.

Example: binary addition

In this example you will add the binary numbers 0011 and 1011. You will use the four rules to help you with your binary addition.

Step 1: **Add the Units column:** Number 1 and Number 2 both have a 1 in the Units column. Rule 3 says 1 + 1 = **10**. Enter the 1 in the carry row of the 2s column. Enter the 0 in the sum row of the Units column.

Step 2: **Add the 2s column:** Number 1 and Number 2 both have a 1 in the 2s column. There is also a 1 in the carry row. Rule 4 says 1 + 1 + 1 = **11**. Enter the first 1 in the carry row of the 4s column. Enter the second 1 in the sum row of the 2s column.

Step 3: **Add the 4s column:** Number 1 and Number 2 both have a 0 in the 4s column. There is a 1 in the carry row. Rule 2 says 0 + 1 = **01**. Enter 0 in the carry row of the 8s column. Enter 1 in the sum row of the 4s column.

Step 4: **Add the 8s column:** Rule 2 says 0 + 1 = **01**. You don't need to enter the 0 because there are no more columns. Enter 1 in the sum row of the 8s column.

Step 1 Rule 3: 1 + 1 = 10

	8s	4s	2s	Units
Number 1	0	0	1	1
Number 2	1	0	1	1
Carry			1	
Sum				0

Step 2 Rule 4: 1 + 1 + 1 = 11

	8s	4s	2s	Units
Number 1	0	0	1	1
Number 2	1	0	1	1
Carry		1	1	
Sum			1	0

Step 3 Rule 2: 0 + 1 = 01

	8s	4s	2s	Units
Number 1	0	0	1	1
Number 2	1	0	1	1
Carry	0	1	1	
Sum		1	1	0

Step 4 Rule 2: 0 + 1 = 01

	8s	4s	2s	Units
Number 1	0	0	1	1
Number 2	1	0	1	1
Carry	0	1	1	
Sum	1	1	1	0

Activity

A student wanted to add 1010 + 0010. They put the numbers into the addition table.

	8s	4s	2s	Units
Number 1	1	0	1	0
Number 2	0	0	1	0
Carry				
Sum				

Copy the table. Use the rules of binary addition to complete the table and find the sum.

Now use the same method to add 0011 + 0111.

Adding bytes

In the example of binary addition in this lesson, you added together two four-bit binary numbers. The method works for binary numbers with any number of bits. You learned in Lesson 1.2 that computers use bytes to store and process data. A byte is 8 bits long. To add bytes together, extend the table used in the example so that it holds 8 bits.

Activity

A student wanted to add two 8-bit binary numbers:

00110111 + 01001010

They extended the addition table to eight columns and put the two numbers into the table.

	128s	64s	32s	16s	8s	4s	2s	Units
Number 1	0	0	1	1	0	1	1	1
Number 2	0	1	0	0	1	0	1	0
Carry								
Sum								

Copy the table. Use the rules of binary addition to complete the table and find the sum.

Now use the same method to add 01011001 + 00001111.

Extra challenge

In this section you carried out a lot of binary calculations. You can check the results for yourself:

- Look at the two binary numbers you added together. Convert them both into decimal numbers.
- Look at the binary number that is the result of the addition. Convert that into a decimal number.

Now you have three decimal numbers. Add the first two numbers together. The result should equal your third number. If it all matches then your binary addition is correct.

Check all your binary addition sums using this method.

Test

1. What is 1 + 1 in binary?
2. What are the four rules of binary addition?
3. Complete this binary addition: 00101101 + 00100101
4. Convert all the numbers in the sum in question 3 to decimal.

1.4 Decimal to binary

In this lesson

You will learn:

- how to convert between decimal and binary
- what an overflow error is.

Convert decimal to binary

When a decimal value is stored in the computer it must be converted to binary. In this lesson you will learn to convert decimal numbers into binary.

Place values

The 1s of a binary number have different values depending on their place in the number. You have learned to show those values as column headings.

Here is an example. It shows the binary number 00100100. The digits have been placed a table with all the column headings.

128s	64s	32s	16s	8s	4s	2s	Units
0	0	1	0	0	1	0	0

You can find the value of each 1 by looking at the column headings. The first 1 is in the 32s column. It has the value 32. The second 1 is in the 4s column. It has the value 4. You can find the total value of the number by adding up the place values. In this case, 32 + 4 = 36.

So the binary number 00100100 has the decimal value 36.

Convert decimal to binary

You can use the same number table to convert any decimal number to binary.

- Start at the left of the table.
- Subtract the place values from the decimal number (without going below 0).
- Whenever you subtract a value put a 1 into the table.
- Put 0 in all the other columns.

Worked example 1

Here is a worked example. You will convert the decimal number 20 into binary.

First draw the grid.

128s	64s	32s	16s	8s	4s	2s	Units

Start at the left. The first value is 128. That is too big to subtract from 20. So put a 0. Count along until you find a value that you can subtract. That is 16.

Put 1 into the 16s column.

128s	64s	32s	16s	8s	4s	2s	Units
0	0	0	1				

Subtract 16 from 20, and there is 4 left. Put 1 into the 4s column.

128s	64s	32s	16s	8s	4s	2s	Units
0	0	0	1	0	1		

You have now got 0 left so fill up the rest of the table with 0s.

128s	64s	32s	16s	8s	4s	2s	Units
0	0	0	1	0	1	0	0

So 20 in decimal is 00010100 in binary.

Worked example 2

Convert the decimal number 161 into binary.

Counting from the left we can subtract values like this:

165 – 128 = 37

37 – 32 = 5

5 – 4 = 1

1 – 1 = 0

The numbers subtracted are 128, 32, 4 and 1. Put 1s into these columns.

128s	64s	32s	16s	8s	4s	2s	Units
1		1			1		1

Put 0s in the other columns.

128s	64s	32s	16s	8s	4s	2s	Units
1	0	1	0	0	1	0	1

So 161 in decimal is 10100101 in binary.

Activity

Convert the following decimal numbers to binary. Show your working.

a 32

b 80

c 69

d 133

The largest number

In this unit you have learned to make numbers with up to 8 bits. That is 1 byte.

- The smallest number you can make with 8 bits is 00000000. That has the decimal value 0.
- The biggest number you can make with 8 bits is 11111111.

What is the decimal value of this number? You can find out by putting it into the binary grid.

128s	64s	32s	16s	8s	4s	2s	Units
1	1	1	1	1	1	1	1

128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255

So the biggest decimal number you can make with 8 bits has the value 255.

What about other numbers?

In real life the computer needs to store numbers outside of this range.
For example:

- numbers bigger than 255
- negative numbers (smaller than 0)
- fractions and decimal values, such as 4.5.

The computer needs to use more than 8 bits (one byte) to store these numbers.
In this unit you will only work with numbers from 0 to 255.

Activity

Which of these decimal numbers **cannot** be turned into a binary number using 8 bits?

a 99

b 222

c 260

d 499

Overflow error

If you try to store a number bigger than 255 using 8 bits then you will get an error. This is called an overflow error. In Lesson 1.3 you learned about binary addition. Sometimes the result of a binary addition will be a number that is bigger than 255. In this case you will get an overflow error.

Worked example

Add the binary numbers 10101000 + 01100001.

Put them into the addition grid. Carry out the sum using the rules of binary addition.

	128s	64s	32s	16s	8s	4s	2s	Units
Number 1	1	0	1	0	1	0	0	0
Number 2	0	1	1	0	0	0	0	1
Carry	1	1	0	0	0	0	0	0
Sum	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**

The final column is the 128s column. There are two 1s in this column. The rules of binary addition tell us that 1 + 1 = 0 and carry 1. But there is nowhere to put the carry value. There are no more columns.

This causes an overflow error. The result is shown as 00001001. That is **not** the right answer to the sum.

Extra challenge

Add the two binary numbers 11001011 + 00111111. Show the overflow error.

Test

1. Convert the decimal number 67 to binary.
2. What is the largest decimal number that can be shown using 8 bits?
3. Complete the following binary addition, showing the overflow error.

   01011100 + 11001000
4. Convert the following decimal numbers into binary and show that adding them together causes an overflow error.

   150 + 120